M13-C3 — Probe catalogue (H1–H7) motivated by closed H0

H0 closed; H1–H4 run/analysed;
H5 deferred; H6/H7 signposted only (not claims)

1 Status

Follow-up probes **motivated by closed baseline H0**, and **not** part of H0 itself. H0 is the completed six-cell AAMAS+ECAI investigation on fixture `m13_bucket3_binary_collider_and` / `n30_seed42`, documented in `learning_analysis.md` / `.tex` and summarised in `experiment.md`.

- H0: **closed** (see `learning_analysis.md`; not redefined here)
- H1: **run / analysed** — nested-prefix AAMAS ablation on `n25_seed42`. Record: `h1_support_ablation.md` / `.tex`. **Not a Bucket 3 claim.**
- H2: **run / analysed** — isolated-D AAMAS target c retains irrelevant d. Record: `h2_irrelevant_covariate.md` / `.tex`. **Not a Bucket 3 claim.**
- H3: **run / analysed** — BK-leading isolated a distracts ECAI target c from BD AND. Record: `h3_bk_leading_distractor.md` / `.tex`. **Not a Bucket 3 claim.**
- H4: **run / analysed** — repository `baseline_cautious` blocks brave root residual α-gadget on the H0 table. Record: `h4_cautious_vs_brave.md` / `.tex`. **Not a Bucket 3 claim.**
- H5: **deferred theory/design question** / not configured / not run / not a Bucket 3 claim — cautious Greedy ABA Learning is separate from H4
- H6 / H7: **signposted only** (prompted by H4 traces; not approved runs)
- **Not** approved Bucket 3 claims
- **Not** locked Bucket 1/2 claim content
- **Not** report-facing prose (`docs/report/findings/` is out of scope)
- Markdown twin: `future_probes.md`

Samuel decision (2026-08-01) — H1 / H2.

1. **H1 approved** — nested-prefix / missing-$(0, 0, 0)$ AAMAS ablation on this fixture (e.g. $n = 25$, seed 42, targets a and b) against the approved $n = 30$ baseline.
2. **H2 design approved** — separate schema-2 fixture with isolated $D \sim \text{Bernoulli}(1/2)$ (variables `a`,`b`,`c`,`d`).
3. **H2c in scope** for the first H2 write-up.

Samuel decision (2026-08-01) — H3.

1. Graph/naming sketch **approved**.
2. Root parameters **approved:** $B \sim \text{Bernoulli}(4/5)$ ($P(B{=}1) = 4/5$, $P(B{=}0) = 1/5$); $D \sim \text{Bernoulli}(7/10)$ ($P(D{=}1) = 7/10$, $P(D{=}0) = 3/10$); $A \sim \text{Bernoulli}(1/2)$ as previously sketched.

3. Fixture id **locked:** `m13_bucket3_binary_bd_and_lead_a` (AND of parents `b`,`d`; BK-leading isolated `a`). Distinct from `m13_bucket3_binary_collider_and` and from the completed H2 fixture id `m13_bucket3_binary_collider_and_iso_d`.
4. **AAMAS contrast in scope** for the H3 write-up (secondary to primary ECAI inspection of target c; valuable as a related but distinct check from H2).

Samuel decision (2026-08-03) — H4 / H5.

1. Add a repository-baseline cautious target-wise configuration, named `baseline_cautious`, which explicitly pins the existing engine defaults and keeps `check_ic` for output generation.
2. Permit explicit `brave` and `cautious` modes in the target-wise runner while leaving the published ECAI and AAMAS configurations unchanged.
3. Prepare H4 on the closed H0 fixture and sample, with all targets generated automatically; roots (a)/(b) are primary and child (c) is a control.
4. Do **not** run H4 as part of the infrastructure change.
5. Defer any cautious Greedy ABA Learning design to **H5**. No `aamas_cautious` configuration is approved.

H1–H4 have been run and analysed (`h1_support_ablation.md`, `h2_irrelevant_covariate.md`, `h3_bk_leading_distractor.md`, `h4_cautious_vs_brave.md`). Approvals do not create a Bucket 3 claim or a run matrix.

2 Motivation (verified facts from the AAMAS arm of closed H0)

On the approved frozen sample `n30_seed42`, configuration `aamas2025` (greedy / `folding_selection(mgr)` / `folding_space(bk)` / brave / `check_ic`):

Target	Outcome	Delta (if any)
`c`	`solved`	`c(A) :- a_val_1(A), b_val_1(A).` (assumption-free)
`a`	`completed_no_solution`	empty
`b`	`completed_no_solution`	empty

Sources:

- collection summary: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/summary.md`
- target c delta: `.../cells/target-c/output/delta.aba`
- target a/b reports and `prolog.stdout` under `.../cells/target-{a,b}/`

For target c, the learned delta string coincides with the evaluator-only positive-state reference in `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/mechanism_reference.json`. Record that as **syntactic coincidence**, not as a causal-recovery verdict.

Procedural reading of the AAMAS root-target path (verified in `.../cells/target-a/output/prolog.stdout`; target b is symmetric):

1. Greedy folding builds **maximal co-occurrence bodies** from BK for each E^+ row (not a minimal-parent search).
2. The both-1 co-occurrence body `a(A) :- b_val_1(A), c_val_1(A)` is accepted under brave entailment.
3. The both-0 co-occurrence body `a(A) :- b_val_0(A), c_val_0(A)` over-generalises; assumption/contrary repair is attempted.

4. Contrary rote learning cites rows **26** and **30** — the only sample atoms $(0, 0, 0)$ in `samples/n30_seed42.csv` — and contrary folding then fails, yielding `completed_no_solution`.

These facts from closed H0 motivate probe families H1 and H2. They do not by themselves approve a Bucket 3 claim.

2.1 Motivation addendum (ECAI arm of closed H0 — for H3)

On the same frozen sample `n30_seed42`, configuration `ecai2024` (`folding_mode(nd)`, `folding_selection(any)`, brave / `check_ic`), target c **solved** with delta (verified):

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

In `.../ecai2024/n30_seed42/cells/target-c/output/prolog.stdout`, the first nd fold on an early E^+ row replaces the sample id by `a_val_1(A)` (the first BK predictor under declared order `a`, `b` for that cell), then assumption repair retains a in the ordinary target body while the second parent enters via a contrary. Collection summary: `.../ecai2024/n30_seed42/summary.md`.

This motivates **H3**: put an *independent* distractor first in BK by naming, under ECAI, on a new fixture. H3 is now **run** / **analysed** (`h3_bk_leading_distractor.md`); it is still **not** a Bucket 3 claim.

3 Ordering

1. **H1** — nested-prefix / missing-$(0, 0, 0)$ ablation on the existing fixture and AAMAS config (seed 42). **Run** / **analysed** (`h1_support_ablation.md`).
2. **H2** — separate schema-2 fixture with isolated covariate D on the $A \to C \leftarrow B$ AND collider. **Run** / **analysed** (`h2_irrelevant_covariate.md`). Lead: AAMAS c retains `d_val_*`.
3. **H3** — separate schema-2 fixture `m13_bucket3_binary_bd_and_lead_a` with BK-leading independent A and AND collider $B \to C \leftarrow D$. **Run** / **analysed** (`h3_bk_leading_distractor.md`). Lead: ECAI first-folds to a and retains a in the theory for c.
4. **H4** — same fixture/`n30_seed42`; locked ECAI brave vs repository `baseline_cautious`; roots a/b primary; c control. **Run** / **analysed** (`h4_cautious_vs_brave.md`). Lead: cautious KO on brave residual α-reuse; roots `completed_no_solution`; control c still solves.
5. **H5 (deferred)** — determine whether and how cautious acceptance should be combined with the Greedy ABA Learning strategy. This is not treated as a one-option configuration change.
6. **H6** / **H7 (signposted only)** — prompted by H4 traces; not approved runs.

Do not expand any family into a broader run matrix beyond these bounded probes without a further Samuel decision.

3.1 Relation among probe families (keep distinct)

Probe	Focus
H1	AAMAS roots; missing negative witness / both-0 over-generalisation on the **current** three-variable fixture
H2	AAMAS target c; irrelevant covariate enters **greedy maximal** co-occurrence bodies (minimality vs solvability); fixture id `m13_bucket3_binary_collider_and_iso_d`
H3	**ECAI** target c (primary); **BK order** puts independent distractor **first**; nd first-fold / repair path; **AAMAS contrast in scope**; id `m13_bucket3_binary_bd_and_lead_a`
H4	**Same fixture/sample**; locked ECAI brave vs repository `baseline_cautious`; roots a/b primary; c control; **run / analysed**
H5	Deferred theory/design question: cautious acceptance with Greedy ABA Learning; no config or run approved
H6 / H7	Signposted only (H4-prompted); length drivers under cautious; brave `asm_intro(sechk)` vs `relto`

H3 is not a duplicate of H2: different DAG, different distractor placement/name, different primary config (ECAI nd vs AAMAS greedy), and distinct fixture ids. The in-scope AAMAS arm on H3 still differs from H2 because the irrelevant variable is BK-leading a and the mechanism parents are b, d.

H4 is not a duplicate of H1 or H3: it holds the table fixed and changes the learning mode while retaining the engine's baseline nd/any/all/relto options. H5 is separate because a cautious Greedy learner needs a paper- and algorithm-level specification before a configuration can be interpreted.

4 Probe family H1 — support sensitivity on AAMAS root targets

Status: **run** / **analysed** / not a claim. Evidence record: `h1_support_ablation.md` / `h1_support_ablation.tex`.

Evaluative stance (locked for H1 reading). Roots have `no_observed_parent_deterministic_rule`. H0 AAMAS root `completed_no_solution` is the **desired** / **correct** outcome. H1 AAMAS root `solved` with co-occurrence Horn rules is an **incorrect** / **spurious** finite-sample solution.

4.1 H1a — spurious root rule under missing negative witness

On this fixture, AAMAS, target a (symmetrically b), seed 42: the nested prefix n=25 **omits all** $(0, 0, 0)$ **rows**. Learning **solved** with a delta that includes the both-0 co-occurrence rule

```
a(A) :- b_val_0(A), c_val_0(A).
```

(respectively `b(A) :- a_val_0(A), c_val_0(A).`), which is **false** as a population implication and is blocked on the closed H0 n=30 table by E^- rows 26 and 30.

Nested-prefix fact (seed 42): rows 1–25 contain no $(0, 0, 0)$ atom; rows 26 and 30 are the only $(0, 0, 0)$ observations in n=30. Verified: `n25_seed42.csv` is a row-for-row prefix of `n30_seed42.csv`.

4.2 H1b — negative witness in the unsafe cell

If the sample contains at least one opposite-label row in that both-0 predictor cell, the both-0 rule fails brave entailment under the observed AAMAS path. On closed H0 n=30 this manifested as `completed_no_solution`, not as emission of only the safe both-1 rule. That H0 outcome is the **correct** reading for roots under the H1 stance.

4.3 Explicit non-claims for H1

- Full support does **not** imply AAMAS will emit a "correct" deterministic root mechanism rule: roots have `no_observed_parent_deterministic_rule` in the evaluator reference.
- "One of each population atom" is a sufficient practical condition on this approved sample, but the operational point is a **label conflict inside the greedy candidate body cell**, not a generic demand for full joint support.
- H1 does not claim that omitting $(0, 0, 0)$ is the only way AAMAS can emit a spurious root rule, nor that every seed behaves identically.
- H1 does **not** claim that root `solved` on n=25 is progress or mechanism recovery.

4.4 Outcome (H1)

Run / analysed. Nested-prefix ablation executed on `m13_bucket3_binary_collider_and` / AAMAS / seed 42 / n=25. Roots a/b **incorrectly** solved with both-1 and both-0 co-occurrence rules. Child c remained AND-solved as a control. Full narrative: `h1_support_ablation.md`.

5 Probe family H2 — irrelevant isolated covariate on AAMAS target c

Status: **run / analysed** / not a claim. Evidence record: `h2_irrelevant_covariate.md` / `h2_irrelevant_covariate.tex`.

Lead finding. On fixture `m13_bucket3_binary_collider_and_iso_d` / `n30_seed42`, AAMAS solves target c with

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_1(A), b_val_1(A), d_val_0(A).
```

so **irrelevant** D **is included**. The evaluator-only reference remains the D-free AND `c(A) :- a_val_1(A), b_val_1(A).`. Failure mode: distractor retention / minimality, not unlearnability.

5.1 H2a — irrelevant covariate enters greedy bodies

Observed as predicted: when both values of D appear among E^+ for c (9 each on n=30), greedy catalogue bodies **include** `d_val_*`.

5.2 H2b — learnability vs minimality

c **still solved** (two D-conditioned rules). Confirms that the interesting failure is retention of an irrelevant predictor, not `completed_no_solution`.

5.3 H2c — sample-edge (also run; not the centre)

Nested n=2 prefix where C=1 realises only d=1: single rule `c :- a_val_1, b_val_1, d_val_1`. Extreme edge. Documented briefly in `h2_irrelevant_covariate.md`. Not the scientific centre of H2.

5.4 Outcome (H2)

Run / analysed. Fixture authored and certified; AAMAS `n30_seed42` primary target c shows distractor inclusion; H2c n=2 also run. Full record: `h2_irrelevant_covariate.md`.

6 Probe family H3 — ECAI first-fold distraction by BK-leading independent variable

Status: **run / analysed** / not a claim. Evidence record: `h3_bk_leading_distractor.md` / `h3_bk_leading_distractor.tex`.

Lead finding. On fixture `m13_bucket3_binary_bd_and_lead_a` / `n30_seed42`, ECAI first-folds to `a_val_*` and retains a in both target rules for c. The emitted delta does **not** coincide with the evaluator-only BD AND `c(A) :- b_val_1(A), d_val_1(A).`. Failure mode: BK-leading distractor inclusion / first-fold distraction, not unlearnability.

6.1 Fixture (as run)

Field	Value
Fixture id	`m13_bucket3_binary_bd_and_lead_a`
Internal / display	`a`,`b`,`c`,`d` / A, B, C, D
DAG	$B \to C \leftarrow D$; **no edges** involving A
A	Bernoulli(1/2); independent of $\{B, C, D\}$
B	Bernoulli(4/5) — $P(B{=}0) = 1/5$, $P(B{=}1) = 4/5$
D	Bernoulli(7/10) — $P(D{=}0) = 3/10$, $P(D{=}1) = 7/10$
C	deterministic $C = B \wedge D$
Evaluator reference for c	`c(A) :- b_val_1(A), d_val_1(A).` (`a` must not appear)
BK order for target c	predictors `a`**, then** `b`**, then** `d` (irrelevant first)
Frozen sample	`n30_seed42`
Primary learning arm	**ECAI** target c
Secondary arm	**AAMAS** on the same frozen sample (**in scope**)

6.2 Scientific question (answered)

When learning target c under **ECAI** on this frozen sample, does BK-leading independent a **distract** the first fold and/or the final ABA shape away from the mechanism-aligned B, D conjunction?

Answer (bounded): yes — first fold selects `a_val_1`; both target rules retain a; metrics body vars $\{a\}$. Secondary: AAMAS also retains `a_val_*` in both Horn bodies (vs H2's trailing-d inclusion).

6.3 Secondary finding (in scope; not the lead)

Brave residual / underdetermining delta on the nested a=1 / α_1–α_3 side. Cross-link H0 ECAI target-a rows 9 vs 26. Reinforces: `solved` + **audit SAT** $\neq$ **mechanism-aligned**.

6.4 Explicit non-claims for H3

- Inclusion of a is **not** causal recovery of an $A \to C$ edge.
- Runner `solved` is **not** automatically correct / mechanism-aligned.
- H3 does not reuse H2's graph; distractor here is A, parents B, D.
- Completing H3 does **not** approve a Bucket 3 claim.

6.5 Outcome (H3)

Run / analysed. Fixture authored and certified; ECAI `n30_seed42` primary target c shows BK-leading a distraction; AAMAS contrast also retains a. Full record: `h3_bk_leading_distractor.md`.

7 Probe family H4 — repository-baseline cautious vs published ECAI brave

Status: **run** / **analysed** / not a claim. Evidence record: `h4_cautious_vs_brave.md` / `h4_cautious_vs_brave.tex`.

Lead finding. On the closed H0 table `m13_bucket3_binary_collider_and` / `n30_seed42`, switching from locked **ECAI brave** to repository `baseline_cautious` **blocks the brave residual-assumption gadget** that had allowed roots a/b to `solved`. Paths coincide through early α_1–α_3 setup; at the closing step `c_alpha_3 :- alpha_2, ...` brave says **OK** and finishes, cautious says **KO**, then exhausts the folding budget and returns `completed_no_solution` (no delta). Control c still solves with a byte-identical delta to locked ECAI c.

7.1 Fixture / arms (as run)

Field	Value
Fixture / sample	`m13_bucket3_binary_collider_and` / `n30_seed42`
Sample hash	`sha256:1edae465...a8ca`
Brave arm	locked `ecai2024`
Cautious arm	`baseline_cautious` (`configs/baseline_cautious_config.pl`)
Primary	roots a, b
Control	c

7.2 Scientific question (answered)

Does repository-baseline cautious learning change the ECAI-brave root outcomes that relied on a residual per-row assumption gadget (rows 9 vs 26), and what happens to control c?

Answer (bounded): yes for roots — cautious KO at the brave closing α-reuse; `completed_no_solution`. Control c still `solved` with identical delta.

7.3 Explicit non-claims for H4

- Not claimed that cautious mode is "better" for causal recovery.
- Not claimed that cautious root no-solution recovers a correct root mechanism (none exists).
- Not a ranking of brave vs cautious as general ABALearn policy.
- Control c string match / `solved` is not causal recovery.
- Under cautious learning, SAT of `bk.sol_chk.asp` is an existential final-artefact integrity witness, not a cautious-consequence proof.

7.4 Outcome (H4)

Run / analysed. Collection at `.../baseline_cautious/n30_seed42/`: $a/b =$ `completed_no_solution`; $c =$ `solved` (identical delta to ECAI). Full record: `h4_cautious_vs_brave.md`.

7.5 Prompted follow-ups (signpost only; not H4 evidence)

H5 remains deferred (cautious Greedy). H6 (length drivers: n, `folding_steps` under fixed cautious) and H7 (brave `asm_intro(sechk)` vs `relto` on the 9/26 conflict) are future probes prompted by H4 traces — not approved runs in this update.

8 Probe family H5 — cautious Greedy ABA Learning

Status: **deferred theory/design question** / not configured / not run / not a Bucket 3 claim.

H5 is deliberately separate from H4. The AAMAS Greedy ABA Learning paper's coherent-case result gives a stratified learned case base with a unique stable model, where brave and cautious acceptance coincide. Its incoherent extension is formulated using brave learning. The target-wise root tasks in the AND collider contain identical predictor configurations with opposite target labels, so they are not a trivial coherent-case application.

For that reason, changing only `learning_mode(brave)` to `learning_mode(cautious)` under the Greedy options would define an exploratory hybrid; it would not by itself establish a paper-backed cautious Greedy method. No `aamas_cautious` config or H5 run is approved. Before H5 infrastructure is considered, the intended learning problem, semantic recovery criterion, and algorithmic interpretation must be specified against the Greedy ABA Learning paper and engine behaviour.

9 Boundary distinctions to preserve

Keep separate throughout any future probe write-up:

Object	Role
AAMAS strategy / greedy maximal co-occurrence search	procedural learner behaviour (H1/H2)
ECAI nd first-fold / BK serialisation order	procedural learner behaviour (H3)
Brave vs cautious entailment / learning mode	procedural / semantic learner behaviour (H4)
Cautious Greedy specification	deferred paper/algorithm question (H5), not an implemented configuration
Finite-sample support and label conflicts in candidate body cells	sample information
Mechanism correspondence for c (AND)	evaluator-only interpretation of a non-root
Absence of deterministic root mechanisms	evaluator reference status
Syntactic delta = reference string	coincidence, not automatic causal verdict
Runner `solved` under brave	brave coverage of $E^{\pm}$, not mechanism recovery

Do not treat predictive rules for roots as mechanism recovery.

10 Cross-links

- Investigation record: `experiment.md`
- Mathematical fixture dossier: `fixture_dossier.tex`
- Bucket 3 planning hub: `../M1.3-bucket3-claims.md` (still **no claim**)
- Research decision log: `docs/research/decisions.md` (2026-08-01 H1–H3 entries; 2026-08-03 H4/H5 entry)
- Fixture pre-run bundle: `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/`
- AAMAS baseline: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/`
- ECAI baseline (H3/H4 motivation): `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/ecai2024/n30_seed42/`
- Repository cautious baseline: `configs/baseline_cautious_config.pl`
- H4 config: `causal/configs/targetwise/m13_bucket3_binary_collider_and/ baseline_cautious/n30_seed42.yaml`
- H4 collection: `causal/outputs/aba_learning/targetwise/ m13_bucket3_binary_collider_and/baseline_cautious/n30_seed42/`
- Target-wise explicit brave/cautious validator: `causal/targetwise/config.py`
- Entailment implementation: `asp_engine.pl` (`entails/5`)

11 Decisions recorded

Item	Decision	Date
H1 nested-prefix / missing-$(0,0,0)$ AAMAS ablation	**Approved** then **run / analysed**	2026-08-01 / 2026-08-02
H2 four-variable isolated-D fixture + AAMAS c	**Approved** then **run / analysed**	2026-08-01 / 2026-08-02
H2c in first H2 write-up	**In scope** (run; brief note only)	2026-08-01 / 2026-08-02
H3 graph/naming; B 80/20, D 70/30; id `m13_bucket3_binary_bd_and_lead_a`; AAMAS contrast	**Design approved** then **run / analysed**	2026-08-01 / 2026-08-02
H4 repository-baseline cautious vs published ECAI brave	**Infrastructure ready** then **run / analysed**	2026-08-03
H5 cautious Greedy ABA Learning	**Deferred theory/design question; no config or run**	2026-08-03
H6 / H7 (H4-prompted)	**Signposted only; not approved runs**	2026-08-03

12 Current status and remaining work

- **H1: complete** (`h1_support_ablation.md`). Not a claim.
- **H2: complete** (`h2_irrelevant_covariate.md`). Lead: AAMAS c retains irrelevant d. Not a claim.
- **H3: complete** (`h3_bk_leading_distractor.md`). Lead: ECAI c first-folds to / retains BK-leading a (not BD AND). Not a claim.
- **H4: complete** (`h4_cautious_vs_brave.md`). Lead: cautious blocks brave root residual α-gadget; roots `completed_no_solution`; control c still solves. Not a claim.
- **H5:** cautious Greedy ABA Learning is deferred pending theory/algorithm specification; no config or run exists.
- **H6 / H7:** signposted only (prompted by H4); not approved for run here.
- Still **no** Bucket 3 claim and **no** expanded run matrix without a further Samuel decision.